LOW-POWER / MASSIVE NETWORKS

Three Roads to Connect IoT

LPWAN · Backscatter · Edge AI & TinyML

A comparative study of low-power, massive IoT communication — with an LPWAN vs. Backscatter simulation

Presentation time: ~20 minutes

김영환: 2021271316 / 김영도: 2024270612 / 최치우: 2025270614

OVERVIEW

What We Will Cover Today

As IoT devices multiply, the core challenge narrows to two questions — how to communicate with very little power, and how to connect a massive number of devices at once.

LPWAN

Long-range, low-power active communication. Connects over kilometers with small data.

Backscatter

Reflects ambient RF signals to communicate. Runs at ultra-low power, even battery-free.

Edge AI / TinyML

Processes data on the device itself. Reduces the data that needs to be sent.

Shared goal: deliver sensor data over a long lifetime, efficiently, where batteries are hard to replace

THE PROBLEM

Why 'Low-Power' and 'Massive' Matter

Limits of the Traditional Approach

Sensor → send everything to cloud → analyze → return result

High power use Transmitting all data drains batteries fast

Network congestion Many devices sending at once cause collisio ns

Latency Round-trip communication hurts real-time response

Rising cost Cost scales with data volume and connections

Two Directions for a Solution

Make transmission cheaper

LPWAN & Backscatter — maximize the power and range effic iency of the link itself

Send less in the first place

Edge AI & TinyML — process on-device and send only what i s needed

→ We surveyed all three, and directly compared the first two in code.

AT A GLANCE

Positioning the Three Technologies

Aspect	LPWAN	Backscatter	Edge AI / TinyML
Role	Long-range comms	Ultra-low-power comms	Low-power processing
Signal	Generates & amplifies	Reflects ambient RF	Computation, not comms
Range	Several km	A few ~ tens of m	Not applicable
Power	100s µW ~ 10s mW	A few µW (~1000× less)	Cuts compute energy
Examples	LoRaWAN · Sigfox · NB-IoT	Ambient Backscatter	TF Lite Micro

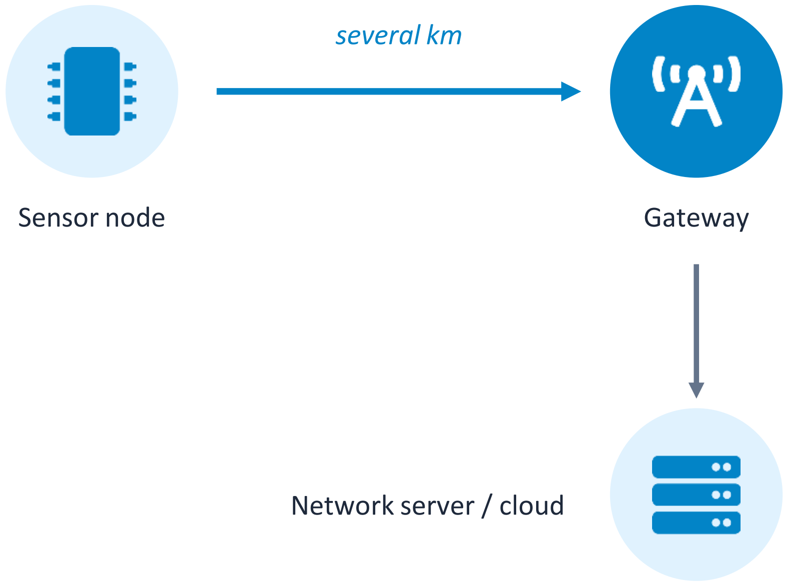
LPWAN — Far, On Little Power

Low-Power Wide-Area Network

An active wireless technology for IoT where batteries are hard to replace and long-range transmission is needed. An internal oscillator and power amplifier generate the RF signal directly.

- ✅ **Overwhelming coverage** Single path loss → several km of range
- ✅ **Infrastructure-independent** Builds its own network with just a base station
- ❌ **High peak power** Generating the signal means large peak power on TX
- ❌ **No streaming** Higher data rate spikes active power — video infeasible

How It Works (one-way long-range link)

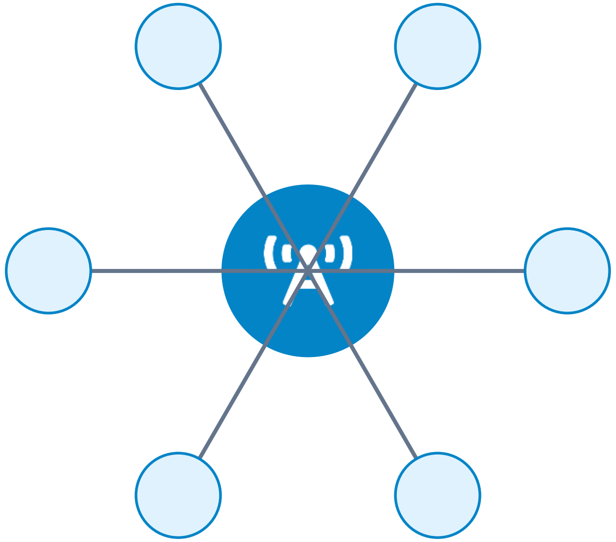


Key: signal travels one way → path-loss exponent ×1

TECHNOLOGY 1 · LPWAN

Topology and Architecture

Main topology: Star / Star-of-Stars



Every node connects to one gateway in a single hop → fast, reliable, and battery-saving

Basic Architecture Layers

- **End Device** — Collects data · responds
- **Access Station / Gateway** — Manages radio link, BER & security
- **Core / Concentrator** — Protocol translation, edge compute
- **LPWAN Server** — Provisioning · registration · routing
- **Application / Cloud** — Database management · big-data analytics

Three Representative Technologies

LoRaWAN

- Star-of-stars topology
- Up to 10-yr battery · ~30 mi rural
- AES-128 security, adaptive data rate (ADR)

Sigfox

- 3D-UNB (time, frequency, space diversity)
- Ultra-light 0–12 byte payload
- High efficiency: 1.8 mJ per useful bit

NB-IoT

- LTE-based, licensed band → high reliability
- PSM & eDRX for 10-yr battery
- 1M+ devices per km² massive scale

Backscatter — It 'Reflects' Signals

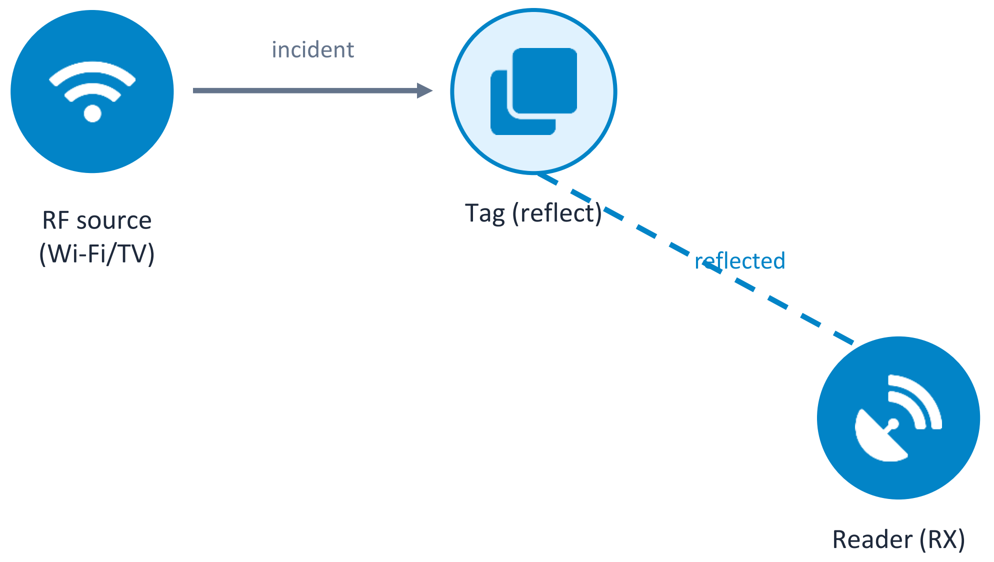
Instead of generating radio waves, it reflects and modulates nearby Wi-Fi, TV, or cellular signals like a mirror. By changing the antenna's impedance to vary reflectivity (load modulation), it creates the 0s and 1s.

Ultra-low power, battery-free
No power-hungry RF generator; drives IC on a few µW, ~1000× more efficient per bit

Reuses existing infrastructure
Uses existing Wi-Fi routers / TV towers as the signal source

Coexists with current comms
Simple reflection avoids interference and diversifies signal paths

How It Works (round-trip dyadic path)



Key: signal makes a round trip → path loss ×2 (r^4 decay)

Strengths · Limits · Applications

Strengths

- ~1000× energy per bit
- 5+ years on one coin cell
- Battery-free / energy harvesting
- Simple circuit even at high freq.

Limits

- Very weak signal
- Range only meters ~ tens of m
- Composite (triple Rayleigh) fading
- Depends on external RF source

Key Techniques

Symbol modulation
BPSK/QPSK via reflection-coefficient control

Non-coherent detection
Decodes from received-energy statistics only

Self-interference cancellation
Essential when TX & RX are simultaneous

Applications

Smart city
Battery-free
smart bins

Smart logistics
Cold-chain
temp/humidity tags

Medical
In-body
biosensors

Agri / environment
Soil moisture
wildfire sensing

KEY COMPARISON

LPWAN vs. Backscatter — The Biggest Difference

	LPWAN	Backscatter
Signal generation	Generates & amplifies (active)	Reflects ambient RF (passive)
Power use	100s µW ~ 10s mW	A few µW (~1000× less)
Path loss	Single (×1)	Round-trip dyadic (×2, r^4)
Range	Several km	A few ~ tens of m
Battery life	Days ~ months on coin cell	5+ yrs / battery-free
Data streaming	Structurally difficult	Favorable even at high rates

→ Need range & independence? LPWAN. Need extreme low power & battery-free? Backscatter.

SIMULATION · DESIGN

How We Compared — Equations & Sources

We computed four metrics under identical conditions in the 868 MHz ISM band. Every equation was verified directly against its primary source (papers, datasheets, standards).

Received power

Log-distance path loss
PL(d)=PL(d0)+10n·log10(d/d0)

Rappaport [1]

Backscatter round-trip

Round-trip path → exponent ×2 (r^4 decay)

Griffin & Durgin [2]

Bit error rate (BER)

BPSK/AWGN
Pb = 0.5·erfc(sqrt(SNR))

Proakis [4]

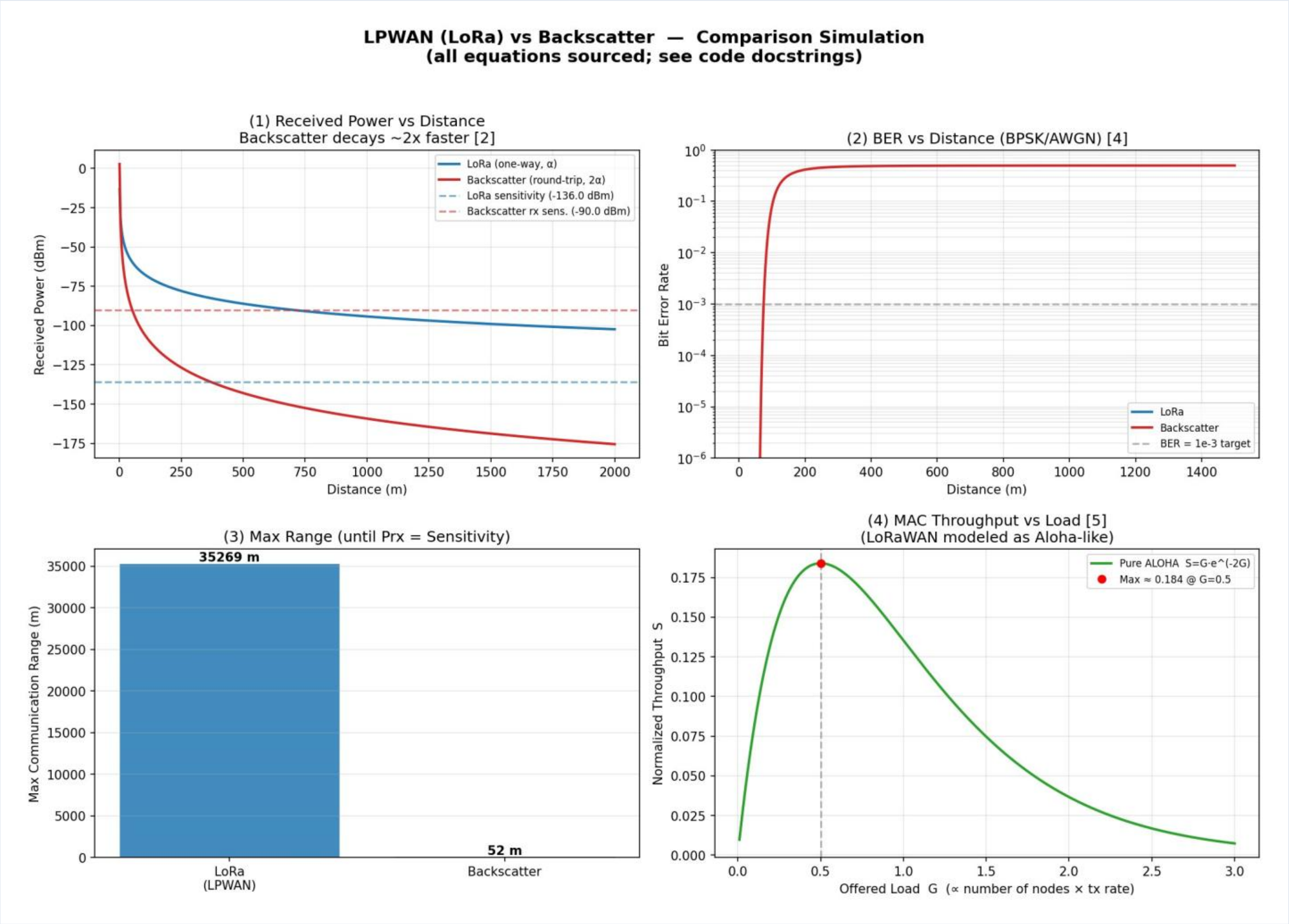
Multi-node throughput

Pure ALOHA
S = G·e^(-2G) (max ≈ 0.184)

Abramson [5]

Key parameters: LoRa TX 14 dBm · sens. −136 dBm | Backscatter carrier 30 dBm · integrated loss 10 dB | path-loss exponent n = 2.7

Simulation Results (Four Metrics)



(1) Received power vs. distance · (2) BER vs. distance · (3) Max range · (4) ALOHA throughput

What the Results Tell Us

35,269 m	52 m	≈ 678×	0.184
LoRa max range	Backscatter max range	Range ratio (LoRa / Back)	ALOHA theoretical max

Interpretation

- Because the round-trip path loss acts twice, Backscatter's received power decays roughly 2× faster with distance.
- As a result, range splits sharply — tens of meters vs. tens of kilometers — the fundamental trade-off between the two.
- BER likewise degrades at much shorter distances for Backscatter (under BPSK/AWGN).
- In the ALOHA model, multi-node throughput peaks at ~18.4% at load G = 0.5 — a shared limit under heavy contention.

Cutting Down the Data to Send

A Shift in How Data Is Handled

Traditional: sensor → send all to cloud → analyze → return

Edge: sensor → analyze on-device → send only needed results

e.g. don't send temperature every second — alert only when a threshold is crossed

Strengths energy efficiency · low latency · less network load · privacy

Limits constrained hardware · needs model compression (quantization/pruning) · possible accuracy loss

Works on top of protocols: MQTT · CoAP · LoRaWAN · NB-IoT

Applications

Smart agriculture
Send only on anomalies

Smart factory
Vibration → predictive maint.

Healthcare
Instant heart-rate / fall alerts

Smart home
Local voice & security

Autonomous driving
Real-time lane / pedestrian

TinyML platform
TF Lite Micro (Google)

Real-World Company Applications

Beyond research concepts, these are shipping in real products: local processing → less data transmitted → faster response.

T Tesla

Autonomous driving system

- Processes camera & sensor data inside the vehicle
- Lane recognition, pedestrian & obstacle det ection
- Key: fast decisions with no comms latency

G Google

TinyML platform

- Provides TensorFlow Lite for Microcontrolle rs
- Runs lightweight AI models even on MCUs
- Key: brings AI to low-power IoT devices

S Samsung

SmartThings smart home

- Connects & automates smart TVs, applianc es, security
- Processes sensors & commands locally for r esponsiveness
- Key: less cloud dependence & network load

In short: Tesla = fast on-device decisions · Google = TinyML tooling · Samsung = smart-home deployment

SYNTHESIS

How the Three Fit Together

These are not competitors — each solves the same goal (low-power, massive) at a different point.

LPWAN

Long-range low-power comms

Wide-area sensor nets, periodic small data

→

Ambient Backscatter

Ultra-low-power comms

Battery-free short-range sensors & tags

→

Edge AI / TinyML

Low-power data processing

Selects on-device which data to send

Ideal system: Edge AI cuts the data → LPWAN/Backscatter transmits it efficiently → low-power massive IoT

CONCLUSION

Wrapping Up

- LPWAN's strength is range and infrastructure independence; Backscatter's is extreme low power and battery-free opera tion.
- Our simulation quantified the trade-off: 35 km vs. 52 m range, path loss ×1 vs. ×2.
- Edge AI/TinyML is not a transport — it cuts the data volume itself, complementing the other two.
- Used together, the three bring us closest to the shared goal of low-power, massive IoT.

Thank you — questions are welcome.

References on the next slide (IEEE format).

REFERENCES

References (IEEE Format)

[1] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2002, Sec. 3.11.3, Eq. (3.93).

[2] J. D. Griffin and G. D. Durgin, “Complete link budgets for backscatter-radio and RFID systems,” IEEE Antennas Propag. Mag., vol. 51, no. 2, pp. 11–25, Apr. 2009.

[3] Semtech Corp., “SX1276/77/78/79 — 137 MHz to 1020 MHz Low Power Long Range Transceiver,” Datasheet, Rev. 7, Camarillo, CA, USA, May 2020.

[4] J. G. Proakis and M. Salehi, Digital Communications, 5th ed. New York, NY, USA: McGraw-Hill, 2008, Eq. (4.3-13).

[5] N. Abramson, “The ALOHA system: Another alternative for computer communications,” in Proc. AFIPS Fall Joint Comput. Conf., vol. 37, 1970, pp. 281–285.

[6] C. Röhrig, D. Heß, and B. H. D. Trinh, “System design for distributed energy management using multiple LPWAN technologies,” in Proc. IECON 2025 — 51st Annu. Conf. IEEE Ind. Electron. Soc., 2025.

[7] Recommendation ITU-R P.525-3, “Calculation of free-space attenuation,” ITU-R, Geneva, Switzerland, Sep. 2016, Annex 1, Eq. (3).

[8] B. S. Chaudhari and M. Zennaro, Eds., LPWAN Technologies for IoT and M2M Applications. Cambridge, MA, USA: Academic Press, 2020.

[9] V. Liu, A. Parks, V. Talla, S. Gollakota, D. Wetherall, and J. R. Smith, “Ambient backscatter: Wireless communication out of thin air,” in Proc. ACM SIGCOMM, 2013, pp. 39–50.

[10] V. Talla, M. Hessar, B. Kellogg, A. Najafi, J. R. Smith, and S. Gollakota, “LoRa backscatter: Enabling the vision of ubiquitous connectivity,” Proc. ACM Interact. Mobile Wearable Ubiquitous Technol. (IMWUT), vol. 1, no. 3, pp. 1–24, Sep. 2017.

[11] D. Darsena, G. Gelli, and F. Verde, “Modeling and performance analysis of wireless networks with ambient backscatter devices,” IEEE Trans. Commun., vol. 65, no. 4, pp. 1797–1814, Apr. 2017.

[12] M. A. Jamshed, B. Haq, M. A. Mohsin, A. Nauman, and H. Yanikomeroglu, “Artificial intelligence, ambient backscatter communication and non-terrestrial networks: A 6G commixture,” arXiv, 2023.

[13] P. Warden and D. Situnayake, TinyML: Machine Learning with TensorFlow Lite on Microcontrollers. Sebastopol, CA, USA: O’Reilly Media, 2019.

[14] C. Banbury et al., “Benchmarking TinyML systems: Challenges and directions,” arXiv:2003.04821, 2021.

[15] Google, “LiteRT for Microcontrollers (TensorFlow Lite for Microcontrollers),” 2024. [Online]. Available: https://ai.google.dev/edge/litert/microcontrollers/overview

Sources [1]–[7] underpin the simulation; [8]–[15] come from the team’s technology surveys (LPWAN, Backscatter, Edge AI/TinyML).